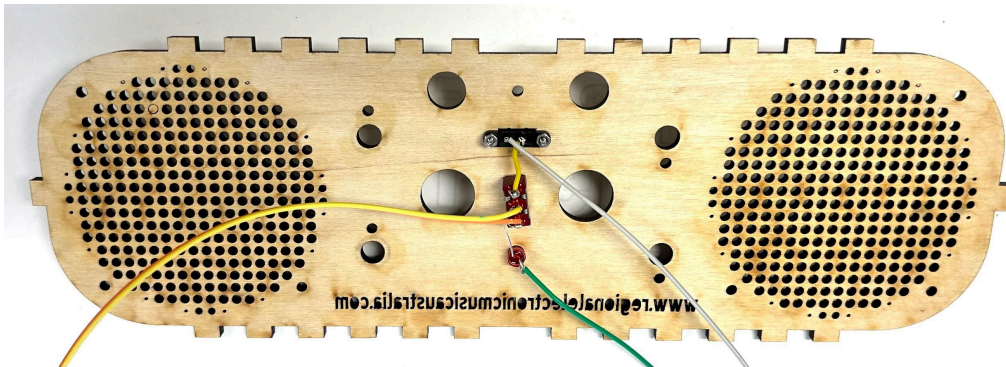
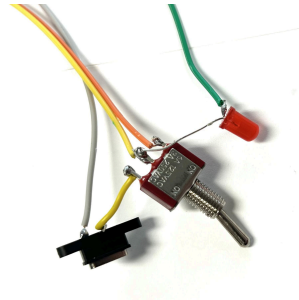


LoFi Granulator - Assembly Guide

Step 1

Attached the power circuit.

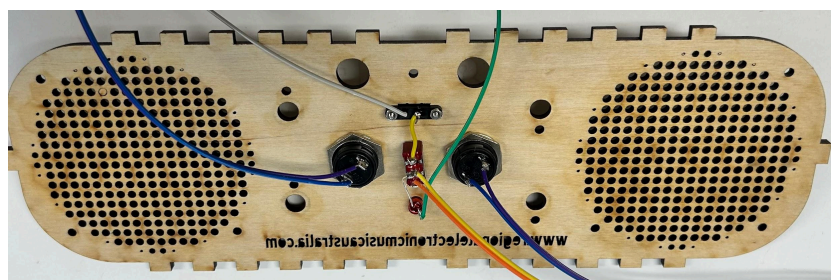
Insert the USB, switch and LED to the centre of the panel. Push in the LED (we'll hot glue it later), bolt in the power switch, and use 2 x M2 bolts and nuts (on the underside) to hold the USB power socket in place.



Step 2

Add buttons on the front.

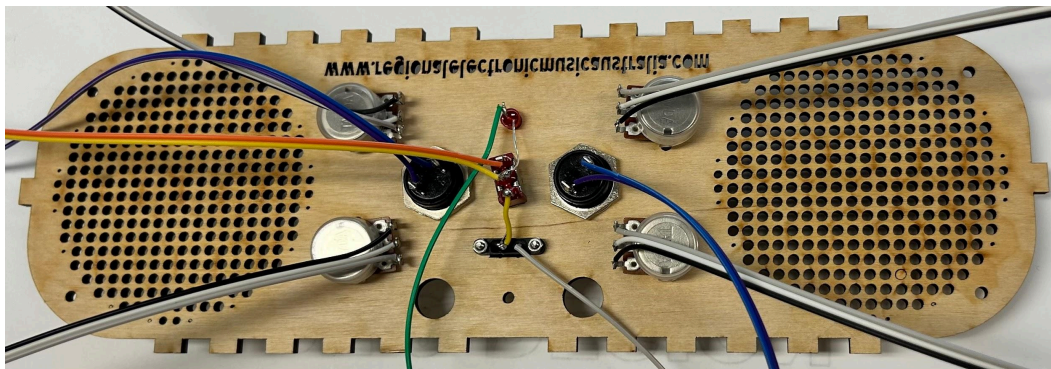
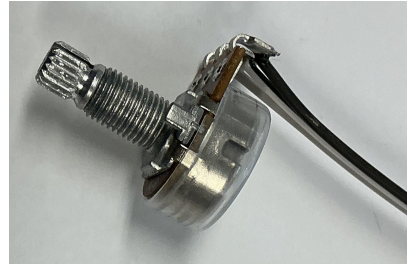
Take off the nuts and washer, insert the buttons in the holes, and secure with the washer and nuts on the back.



Step 3

Add the potentiometers.

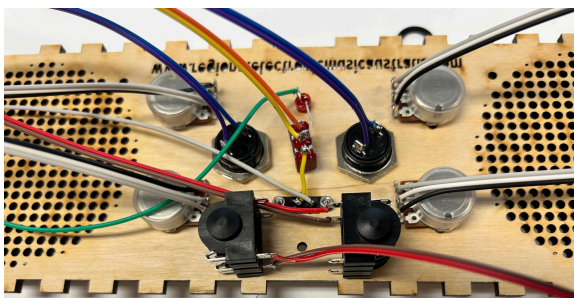
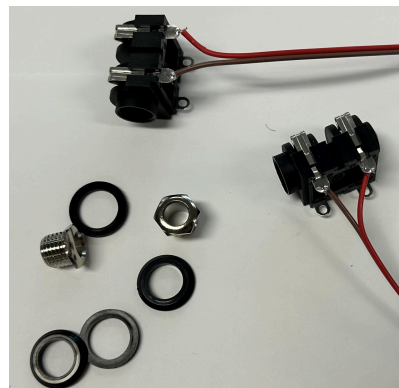
Insert the pots through the holes, make sure they are correctly oriented (see photo), add a washer and nut to the top, and tighten in place.



Step 4

Add the jacks.

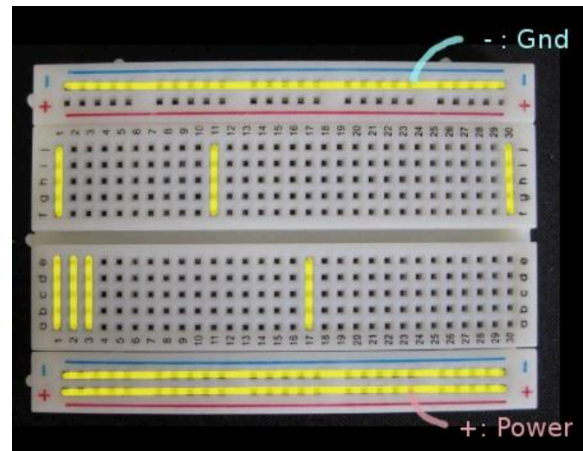
Screw off the front nuts, attach the jacks to the faceplate, and reapply screws to the outside.



Step 5

Stick down the breadboard

The electronic breadboard allows us to assemble components without soldering. Holes are connected within the inner columns and outside rows (rails) so that components inserted into the same column or rail are 'wired' together. The columns are divided into two sets by the 'gap' that runs down the centre of the breadboard.



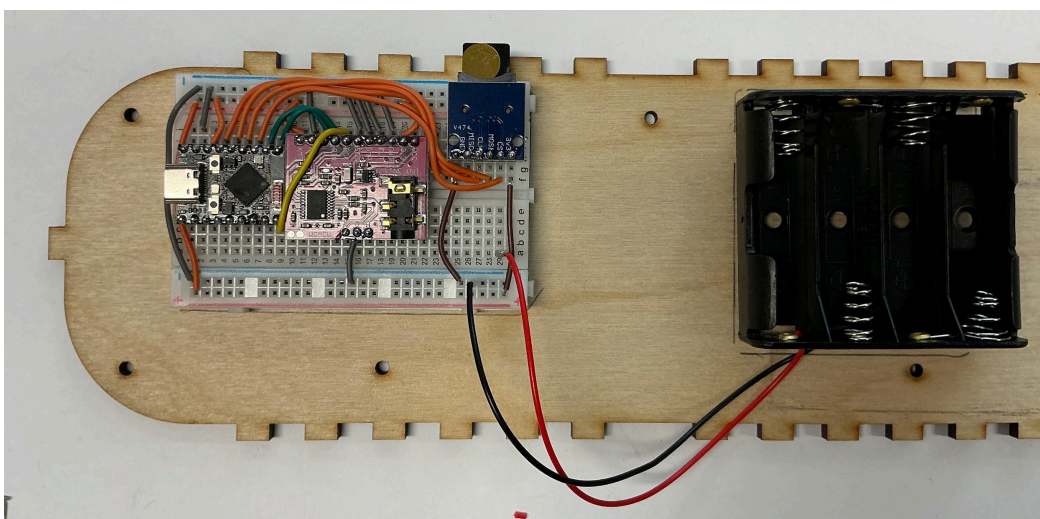
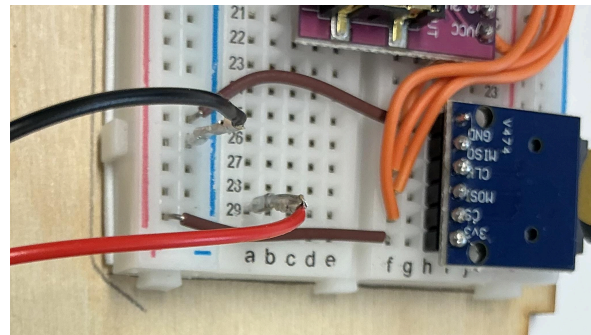
Step 6

Attach the battery holder.

Stick down the battery pack.

Connect the red wire to an empty row on the left side (a - e) of the breadboard channel. We used column A, row 29.

Connect the black wire to the ground row.



Step 7

Connect the switch.

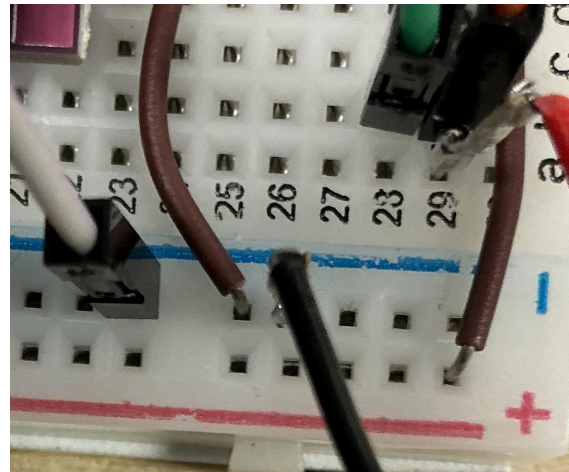
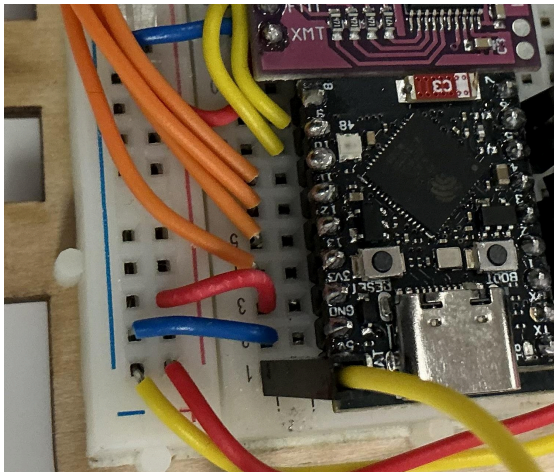
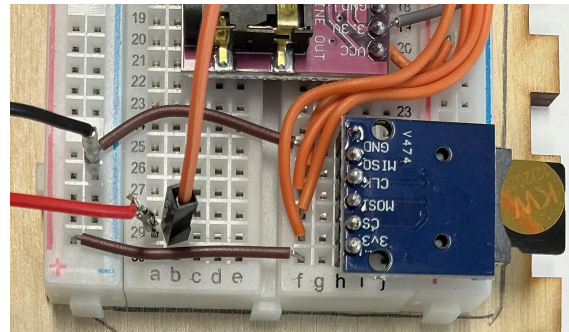
There is an orange and yellow wire coming from the switch.

Connect the orange wire to column I, row 29

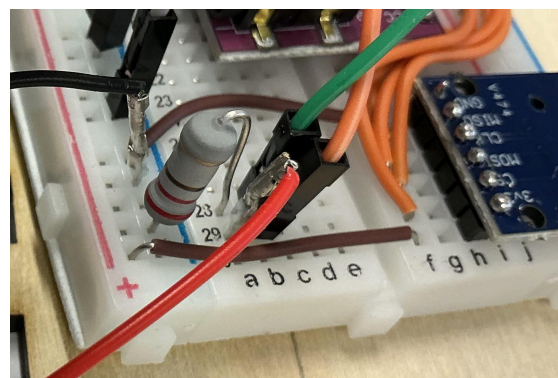
Connect the yellow wire to column C, row 1

Connect the green wire attached to the LED to row 28, column B

Connect the grey wire from the USB to GND (-)



Place the resistor between gnd (-) and row 28 column A



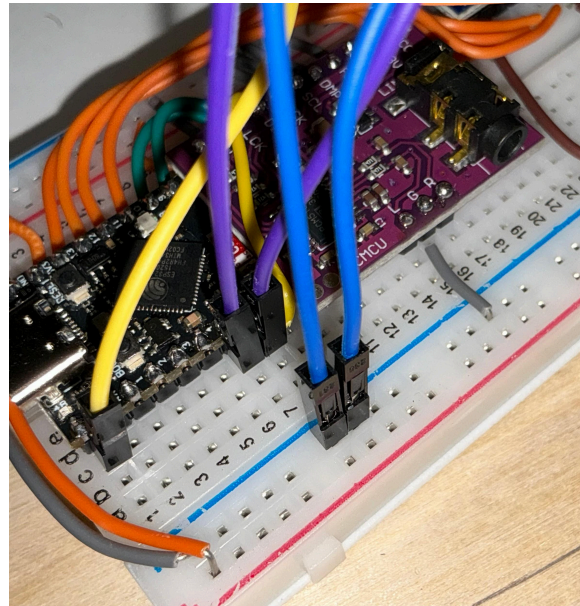
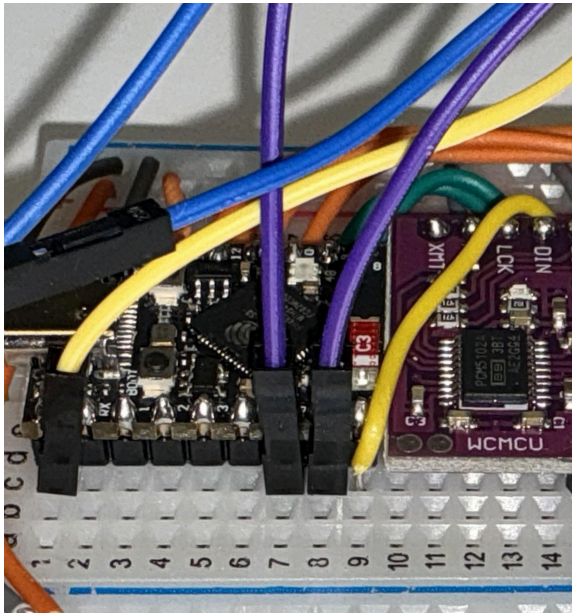
Step 8

Attach the button wires to the breadboard

The purple wire of the Green button goes to row 8, column C

The purple wire from the Blue button goes to row 7, column C

The blue wire of each button goes to the ground (-)



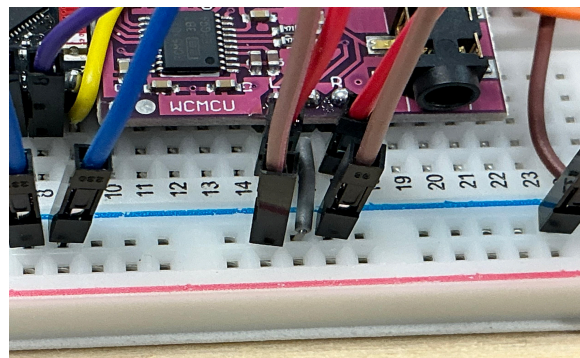
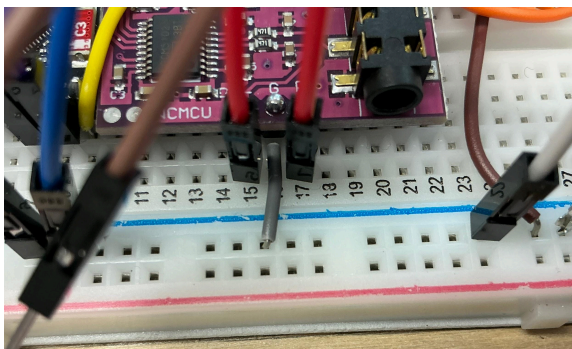
Step 9

Add the audio jacks

Attach the red wire from the left jack to Row 15, column B

Attach the red wire from the right jack to Row 17, column B

Attach the brown wires from the jacks to ground.



Step 10

Attach White wires

L&R is from front view

We use column c for most of these but you can use any column a, b or c as long as the row is correct

Btm L - row 3, column c

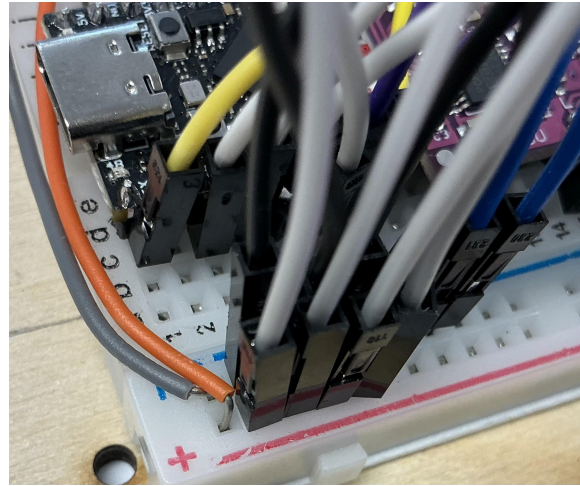
Top L - row 4, column c

Top R - row 5, column c

Btm R - row 6, column b (I used b as there was no room in c)

The black wires go to gnd (-)

The grey wires got to +



Step 11

Add the resistor

There is a resistor between the power the red LED.

Step 12

The amplifier

Connect the blue cable to RIN, the green cable to GND, and the yellow cable to LIN

Speaker 1

- Connect the purple cable from the speaker to ROUT+
- Connect the grey cable from the same speaker to ROUT-

Speaker 2

- Connect the black cable from the speaker to LOUT+
- Connect the white cable from the same speaker to LOUT-

Use the double-sided tape to stick the amplifier board to the base

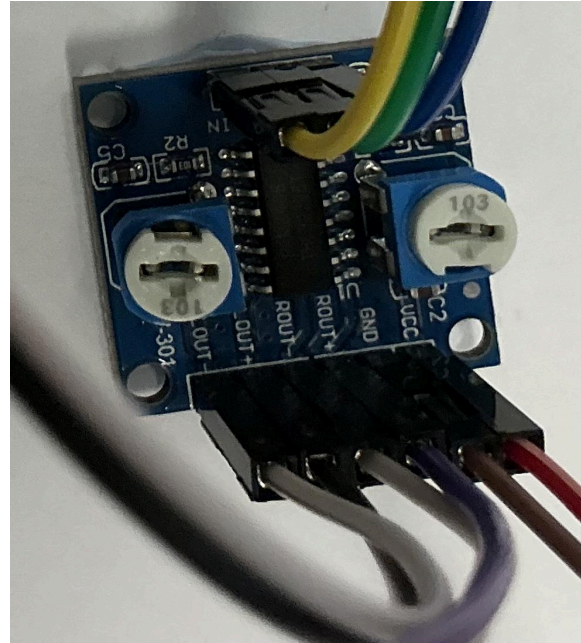
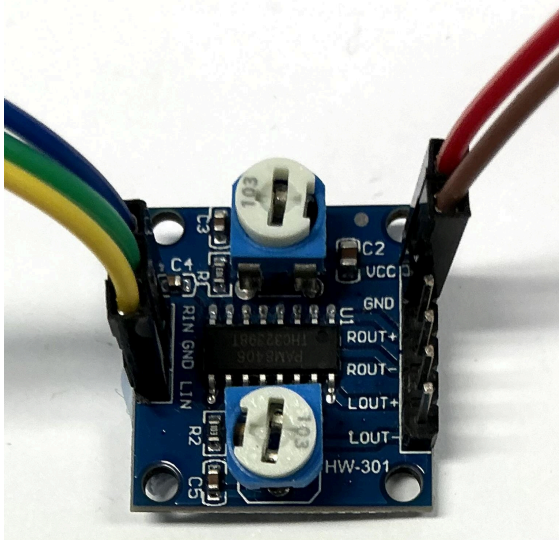
Amp Power

- Connect the other end of the red wire to +

- Connect the other end of the brown wire to GND (-)

Audio In

- Connect the yellow wire to row 15, column A
- Connect the blue wire to row 1,7 column A
- Connect the middle green wire to ground



Step 13

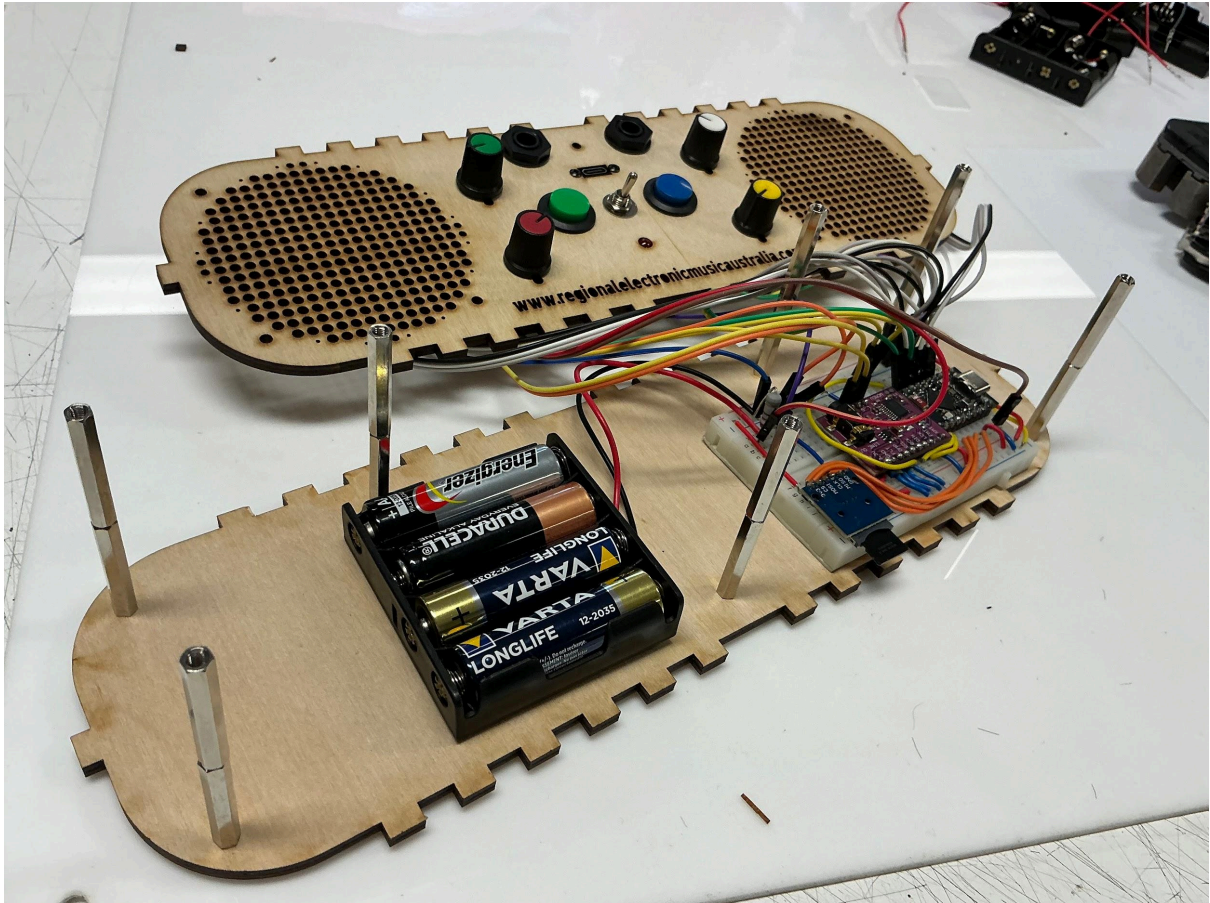
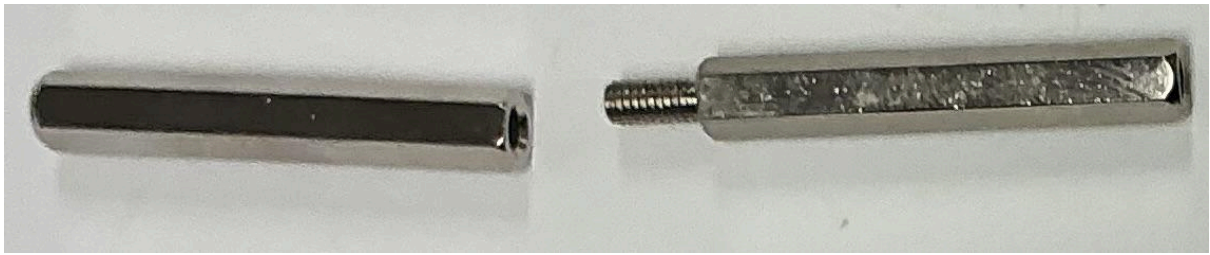
Testing

- Insert an SD card with WAV files, Power the granulator, and press the Green button to hear a sound.
- Adjust the amplifier volume with the two screws on the amp board.

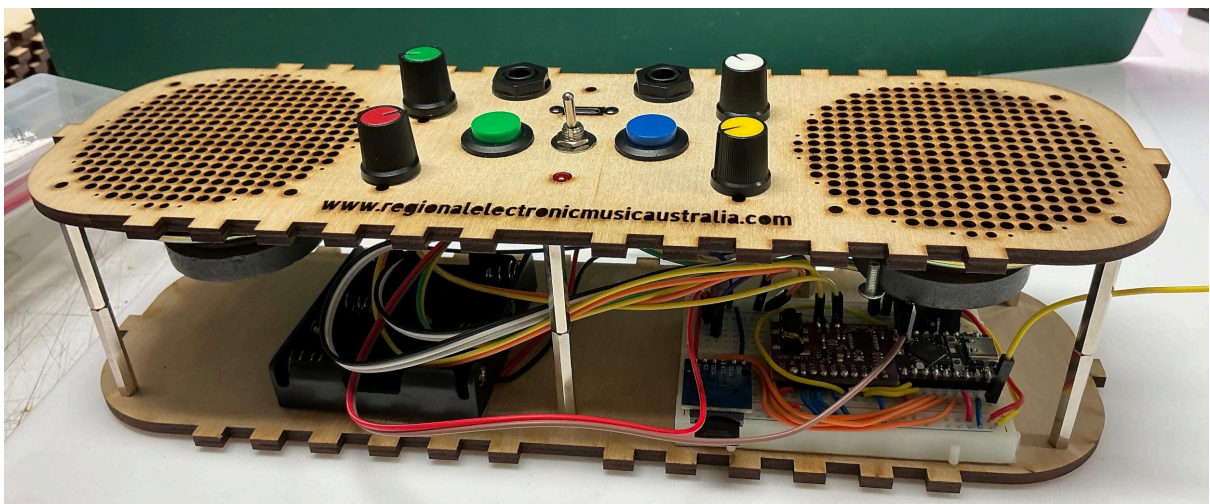
Step 14

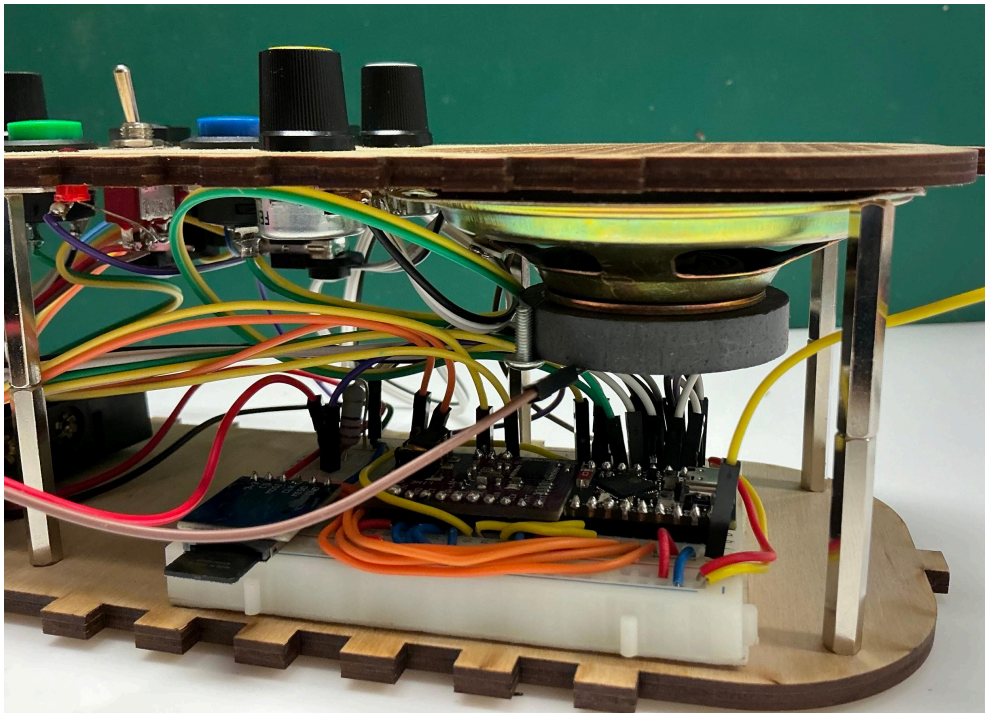
Assemble the Case

- Add the standoffs to the base
- Connect the two types of standoffs together and screw them to the base

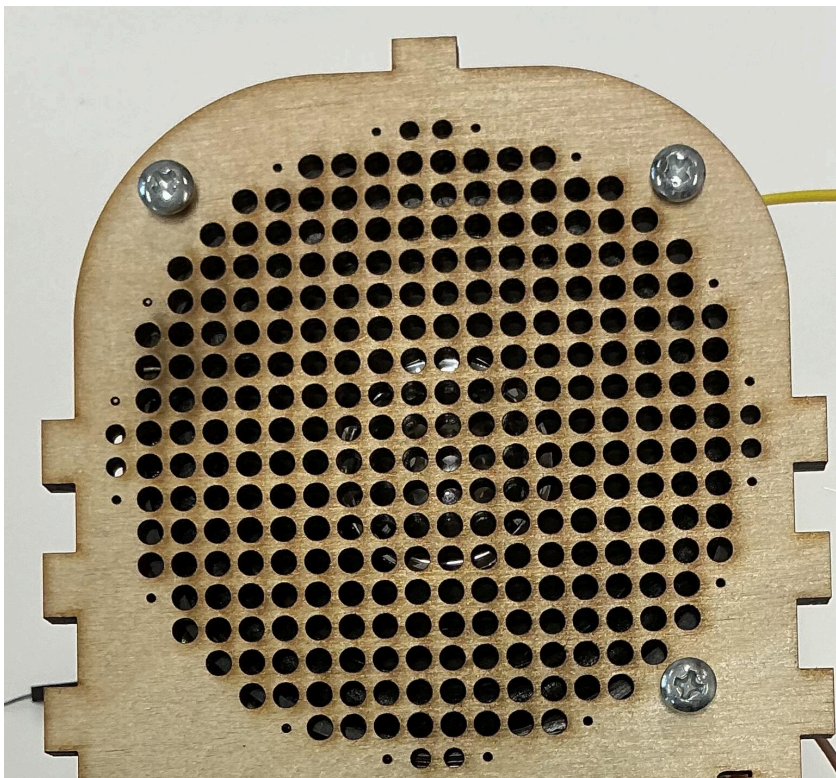


- Place the speakers on the standoffs





- Screw the speakers between the standoffs and the lid



- Add the side of the box
- Add rubber feet to the box